Course AE4-135 Rotor and Wake Aerodynamics

Assignment III: Unsteady Aerodynamics of Airfoils

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TU Delft

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1
Introduction

1.1. Abstract

In this report, we build and benchmark an unsteady Discrete Vortex Model of thin lifting airfoils. This report includes the fundamental background on Lumped-Vortex Element theory, as well as the demonstrations of fundamental quantities for analyzing the aerodynamic performance of thin airfoils modeled as flat plates. Here, we analyze the lifting force coefficient, the velocity and pressure fields, and the shed vortex patterns formed in the wake for various steady and unsteady flow cases. The assessed test cases consist of i. steady flows at different angles of attack; ii. unsteady, plunging flows at different normalized plunging frequencies; and iii. unsteady, pithing flows at different normalized pithing frequencies.

1.2. Background

1.2.1. The Lumped-Vortex Method

A potential flow ($\nabla \cdot (\nabla \times \vec{u}) = 0 \mid \vec{u} = u_x\hat{i} + u_y\hat{j} + u_z\hat{k}$) flow field around an airfoil can be generally decomposed into a (far-field) potential flow and a (near-wall) boundary layer flow. At the boundary layer flow, viscous flow effects are dominant and produce large velocity-field gradients which, in their turn, yield high-vorticity velocity fields, i.e., $\vec{\omega} = \nabla \times \vec{u} \gg 0$. The far-field flow, on the other hand, is typically free of viscous effects and thus can be roughly approximated as an irrotational ($\vec{\omega} = \vec{0}$) flow. Therefore, in the context of potential-flow theory (see Chapter 3 of [1]), the flowfield around an arbitrary two-dimensional blade can be described in terms of the distribution of source flows, $\nabla \cdot \vec{u}$, and vorticity, $\vec{\omega}$, such as illustrated in the top panel of Fig. 1.1.

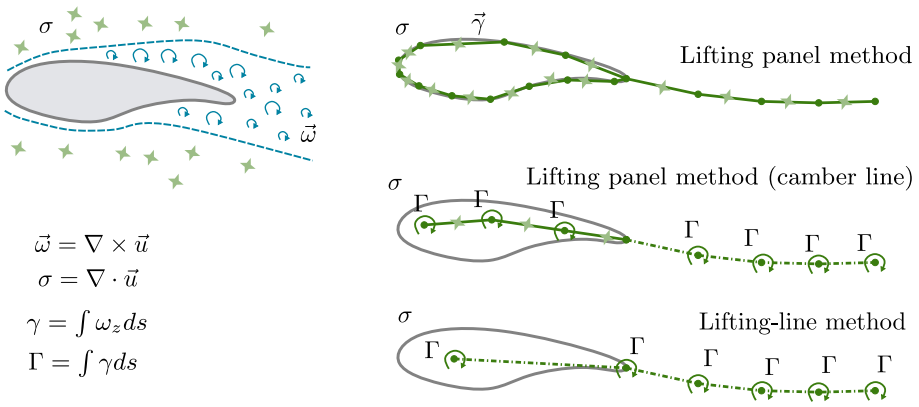


Figure 1.1: Flowfield approximation from three-dimensional to Lifting-line method.

The discrete vortex model presented here is based on the Lumped-Vortex Element methodology and serves for solving numerically the integral of vorticity (see γ Fig. 1.1). In the current report, we model the dynamics of an airfoil based on thin-airfoil assumptions, i.e., the airfoil is approximated by line segments, referred to as panels, oriented along its camber line. In this methodology (see "Lifting panel method (camber line)" in Fig. 1.1), we ignore the boundary-layer dislocation effects, and only the flow effects due to the airfoil's curvature are modeled. The advantage of such an approach is that the

boundary conditions are specified on the airfoil's camber line without the need for the small-disturbance approximation [2].

Let us assume, for simplicity, that the airfoil can be modeled as a flat plate of chord length c. If we define the vortex point, of vortex strength Γ, on the center of pressure of the plate, i.e., at the quarter-chord length of the plate, according to the Kutta–Joukowski theorem, this concentrated vortex yields a lifting force per unit span of:

$$L' = \frac{1}{2}\rho U_\infty^2 \Gamma \tag{1.1}$$

where ρ is the fluid's density, and $U_\infty^2 = \vec{U} \cdot \vec{U}$, where $\vec{U}$ is the airfoil's velocity (kinematic velocity) in an inertial reference frame.

If the flat plate is to be represented by a single vortex of strength Γ, the "non-entry" boundary conditions requiring zero normal flow at the surface of this plate can be specified by a single point too. This point, referred to here as "collocation point", has to be located at the three-quarter chord length of the plate to account for the *Kutta condition* at the trailing edge (see page 115 of [2]). This notion of vortex and collocation points can be extrapolated to camber lines represented by multiple flat plates, such that we lump the vorticity onto vortex points of each plate and reinforce the non-entry boundary condition at the three-quarter length of each plate, such as illustrated in Fig. 1.2.

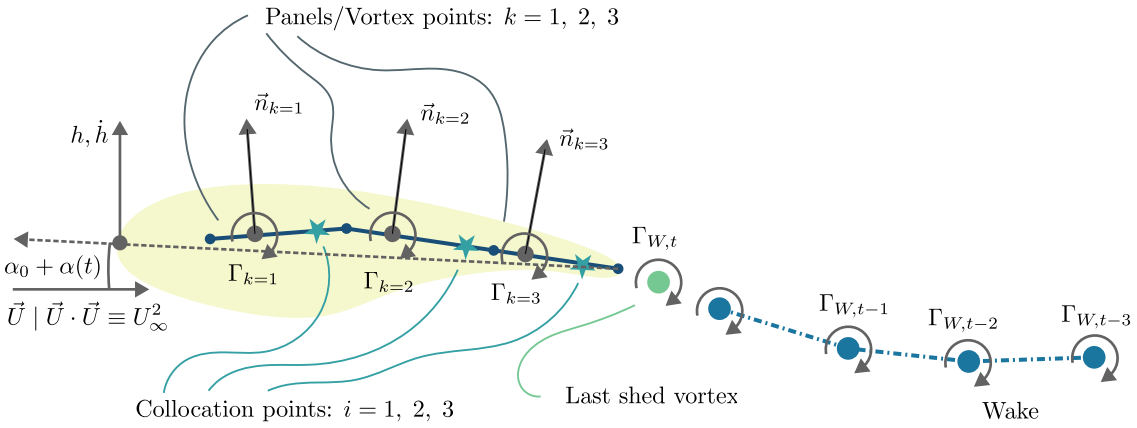


Figure 1.2: Schematic diagram illustrating the elements of the implemented lifting panel method.

1.2.2. Rate of change of Circulation: Kelvin's Theorem

Consider the circulation Γ around a fluid curve in an incompressible inviscid flow with conservative body forces $\vec{f}$. It can be shown that the material derivative of Γ can be expressed as:

$$\frac{D\Gamma}{Dt} = -\oint_C d\left(\frac{p}{\rho}\right) + \oint_C \vec{f} \cdot d\vec{l}, \tag{1.2}$$

where C is an arbitrary curve enclosing a flow region where $\vec{f}$ exists, p is the pressure field, and $d\vec{l}$ is the path describing C. If $\vec{f}$ is conservative, then the second integral on the RHS of the equation above, which described the work done by $\vec{f}$, is identically zero. Moreover, the integral of a perfect differential around a closed path, i.e., the first term on the RHS of the same equation, is also zero. Hence, $D\Gamma/Dt = 0$. This result is a form of angular momentum conservation known as *Kelvin's Theorem* (see page 26 of [2] for a more detailed discussion).

Consider an airfoil at rest at $t = 0$ and constant forward motion at $t > 0$. At $t = 0$, the circulation around the airfoil is $\Gamma_{\text{airfoil}} = 0$. As the airfoil is put to move, a circulation $\Gamma_{\text{airfoil}} \neq 0$ develops around the airfoil. According to Kelvin's theorem, a starting vortex Γ_{wake} must form such that the total circulation around a line surrounding both the airfoil and the wake remains unchanged:

$$\frac{D\Gamma}{Dt} = \frac{\Gamma_{\text{airfoil}} + \Gamma_{\text{wake}}}{\Delta t} = 0. \tag{1.3}$$

The above is only possible if the starting vortex's circulation has an opposite and equal magnitude circulation-strength to the airfoil's. This last result enforces that the circulation of the last shed vortex in the wake, $\Gamma_{W,t}$, plus the total circulation over the airfoil, $\sum_k \Gamma_k$, must add up to the total circulation over the airfoil at the time $t-1$. In the following section, we will explore this property to express a system of equations that allows us to predict the (unknown) vortex strengths of the plates and the last shed vortex from the (known) vortex strengths of the shed vortices.

2
Methodology

2.1. Algorithm and Methods

The implemented VPM approach consists of iteratively solving for the airfoil and latest shed vortex strengths (see Fig. 1.2). Consider an airfoil discretized into $k = 1,\ 3,\ \cdots\ n_P$ panels and a wake discretized into $t = 1,\ 2,\ \cdots,\ n_t$ shed vortices, each corresponding to a vortex shed at the t-th time step ($t \in [0, n_t \Delta t]$). Each panel is associated with a vortex point, $i = 1, 2, \cdots, n_p$, at their quarter-lengths, and with a collocation point $i = 1, 2, \cdots, n_p$, at their three-quarter lengths. At the i-th collocation points, we specify the non-entry boundary conditions, i.e., the velocity component normal to the i-th collocation point must be zero.

The j-th shed vortex induces a velocity $(u, v)_{ij}$ on the i-th collocation point. The normal-velocity component induced on the i-th collocation point is thus given by:

$$a_{ij} = (u, v)_{ij} \cdot \vec{n}_i, \tag{2.1}$$

where $\vec{n}_i$ is the normal direction to the i-th collocation point.

The kinemetic velocity of the airfoil perceived by the i-th collocation point is $\vec{U} = (U_x(t), U_y(t))_i$, whereas the wake-induced velocity perceived by the i-th collocation point is $(u_{W,x}, v_{W,y})_i$. For this discrete-vortex method, the selected singularity is the lumped-vortex element, and its induced velocity, $\vec{u} = (u_x, u_y)$, at a point (x, y) in the flow is given by:

$$\begin{Bmatrix} u_x \\ u_y \end{Bmatrix} = \frac{\Gamma_j}{2\pi r_j^2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{Bmatrix} x - x_j \\ y - y_j \end{Bmatrix} \tag{2.2}$$

where $r_j^2 = (x - x_j)^2 + (y - y_j)^2$.

Notice that we define the "wake" as all the $t - 1$ shed vortexes at the time-step t. This definition is essential since we know the strength $\Gamma_{W,t-1}, \Gamma_{W,t-2}, \cdots \Gamma_{W,0}$ of the $t - 1$ wake vortexes at the t-th time-step, whereas the strength of latest wake vortex, $\Gamma_{W,t}$, and the strength of the airfoil vortexes, Γ_k, are the unknown of the problem at the time-step t.

The non-penetration boundary condition on the panels along the camber line is re-inforced by vanishing the normal velocity component, a_{ij}, at the i-th collocation point:

$$\sum_{k=1}^{k=n_p} (a_{i,k}\Gamma_k) + a_{W,t,i}\Gamma_{W,t} + (U_x(t) + u_{W,x}, U_y(t) + u_{W,y})_i \cdot \vec{n}_i = 0 \tag{2.3}$$

where $a_{W,t,i}$ is the influence coefficient of the latest wake vortex on the i-th collocation point.

Notice that for a rigid airfoil model, we can build a matrix $A_{(n_p+1)\times(n_p+1)} = a_{i,k}$ that relates the normal velocity induced by the k-th panel on the i-th panel, and the normal velocity induced by the latest wake vortex on the i-th panel. Thus, we can build from 2.3 a linear system of the form $[A]\{\Gamma\} = \{\text{RHS}\}$. The i-th row of this linear system is thus given by:

$$\begin{bmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,n_p} & a_{W,t,i} \end{bmatrix} \cdot \begin{Bmatrix} \Gamma_1 & \Gamma_2 & \cdots & \Gamma_{n_p} & \Gamma_{W,t} \end{Bmatrix} = -(U_x(t) + u_{W,x}, U_y(t) + u_{W,y})_i \cdot \vec{n}_i \tag{2.4}$$

Equation 2.4 gives us the first n_p strengths of the airfoil vortexes, Γ_k, with the kinematic and wake-induced velocity perceived at the i-th collocation point. However, we still need to define the last row of the linear system, i.e., the coefficients $a_{n_p+1,k}$ relating the airfoil vortex strength, Γ_k, with the latest shed vortex, $\Gamma_{W,t}$. This last row can be obtained by reinforcing Kelvin's condition, as previously stated in Eq. 1.3:

$$\sum_{k=1}^{k=n_p} \Gamma_k(t) - \sum_{k=1}^{k=n_p} \Gamma_k(t-\Delta t) + \Gamma_{W,t} = 0 \tag{2.5}$$

Thus, the two expressions above can be combined into a single linear system of the form:

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n_p} & a_{W,t,1} \\ a_{2,1} & a_{2,2} & \cdots & a_{1,n_p} & a_{W,t,2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n_p,1} & a_{n_p,2} & \cdots & a_{n_p,n_p} & a_{W,t,n_p} \\ 1 & 1 & \cdots & 1 & 1 \end{bmatrix} \begin{Bmatrix} \Gamma_1 \\ \Gamma_2 \\ \vdots \\ \Gamma_{n_p} \\ \Gamma_{W,t} \end{Bmatrix} = \begin{Bmatrix} \mathrm{RHS}_1 \\ \mathrm{RHS}_2 \\ \vdots \\ \mathrm{RHS}_{n_p} \\ \sum_{k=1}^{k=n_p} \Gamma_k(t-\Delta t) \end{Bmatrix} \tag{2.6}$$

where $\mathrm{RHS}_i = -(U_x(t) + u_{W,x}, U_y(t) + u_{W,y})_i \cdot \vec{n}_i$.

Notice that the first $(n_p + 1) \times n_p$ entries of the matrix A only need to be calculated once for a rigid flat-plate airfoil model. In order to determine the entries $a_{W,t,i}$, however, we need to specify the distance from the shed vortex $\Gamma_{W,t}$ to the flat-plate airfoil. Here, we impose that the location of the latest shed vortex is 80% of the travel distance of the airfoil's trailing edge.

The algorithm follows as:

1. Calculate the entries of the matrix A by solving Eq. 2.2 with $\Gamma_k = 1$:

$$a_{ik} = \frac{\mathrm{RHS}_i}{\Gamma_k = 1}$$

2. Determine kinematic velocity at time-step t, $\vec{U} = (U_x(t), U_y(t))_i$
3. Determine the total wake-induced velocity at the i-th collocation point, $(u_{W,x}, u_{W,y})_i$, by all the shed vortex from times $[t-1, 0]$ using Eq. 2.2.
4. Establish the RHS vector by adding up the wake-induced velocity from step 3 to the airfoil's kinematic velocity, $\vec{U}$, from step 2.
5. Shed a vortex at 80% of the travelled distance of the airfoil's trailing edge.
6. Establish (or update) the last column of the matrix A using:

$$a_{W,t,i} = \frac{\mathrm{RHS}_i}{\Gamma_{W,t} = 1}$$

7. Perform vortex roll-up by moving the shed (wake) vortexes accordingly to the local velocity field using Eq. 2.2.
8. Solve the linear system $[A]\{\Gamma\} = \{\mathrm{RHS}\} \rightarrow \{\Gamma\} = \{\mathrm{RHS}\}[A]^{-1}$.
9. Repeat steps 2-8 for the desired number of time-steps.

Thus, the algorithm consists of moving the airfoil at a specified (kinematic) velocity, shedding a vortex at 80% of the trailing edge's travel distance, and calculating the vortex strengths of both airfoil's vortexes and the latest shed vortex by balancing them out with the strength of the vortex shed in the previous time-steps. Notice also that we use the system's linearity to establish the matrix A by calculating its entries for unitary circulation strengths Γ. Moreover, since only the position of the shed vortex is not known *a priori*, only the last row of the matrix A has to be updated every time step. All other entries can be calculated only once before the start of the time-advancing iterations.

3
Results

This section is subdivided as follows. In 3.1, we determine the sizing of input parameters of the lifting panel method for discretization-independent results. In 3.2, we analyze the lift force coefficient for steady flows at different angles of attack. Finally, in 3.3 and 3.4, we compare lifting force coefficients of pithing and plunging airfoil motions to steady results. We also show vorticity and pressure coefficient fields for the unsteady flows at different reduced frequencies.

3.1. Discretization-independence analysis

In this subsection, we perform the sizing-independence analysis of the input parameters. For this study, results are considered parameter-independent if their magnitude varies less than 1% with the change of the respective parameter. Since we are interested in the unsteady loadings on the flat-plate airfoil, the total circulation over the flat plate:

$$\text{Total airfoil circulation} = \sum_k \Gamma_k,$$

is adopted as a reference parameter for the discretization-independence. Three input parameters are assessed. Namely, the number of panels in the flat-plate airfoil model, n_p, the *wake vortex spacing*, defined here as

$$\text{Wake vortex spacing} = \frac{\Delta t U_\infty}{c}, \tag{3.1}$$

and the *wake length*, defined here as:

$$\text{Wake length} = \frac{U_\infty t_f}{c}. \tag{3.2}$$

In Eqs. 3.1 and 3.2, t_f is the total flow time and Δt is the time-step in seconds.

In Fig. 3.1, we analyze the convergence of the total circulation on the flat-plate airfoil for increasing wake lengths. For this analysis, we kept the number n_p of plates constant and equal to 5. The flow is steady at an angle of attack, α, of $\alpha = 10$ degrees. To isolate the effects of the wake length from the time-step sizing, we scaled the time-step sizing with the wake length, such that Δt is the same in all test cases. Notice from Eq. 2.2 that the induced velocity from a wake vortex on a collocation point reduces with the distance; thus, we expect convergence of the results for a finite wake length. From the results shown in Fig. 3.1, we conclude that a wake extending a time window of $t_f = 25c/U_\infty$ is sufficient for wake-length-independent results.

After establishing the sizing of the wake length for wake-length independent results, we investigate the number of panels n_p necessary for panel-sizing independent results. The wake length was set for the current analysis so that $t_f U_\infty / c = 25$. We test here n_p values ranging in $[5, 30]$. The results of this analysis are shown in Fig. 3.2. From the results shown in the figure, the total airfoil circulation is independent of the number of panels for $n_p > 25$. Notice, however, that changes of Γ_k concerning n_p are negligible for $n_p > 10$. Thus, for the current study, $n_p = 10$ is sufficient for sizing-independent results.

The third and last input parameter for which a sensitivity analysis is carried out is the sizing of the time-stepping, Δt. Notice, however, that there are two alternative non-dimensional characterizations for

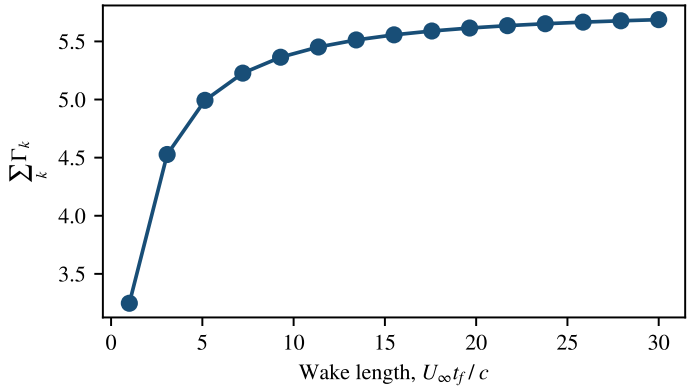


Figure 3.1: Convergnce of total airfoil circulation for increasing values of normalized wake length.

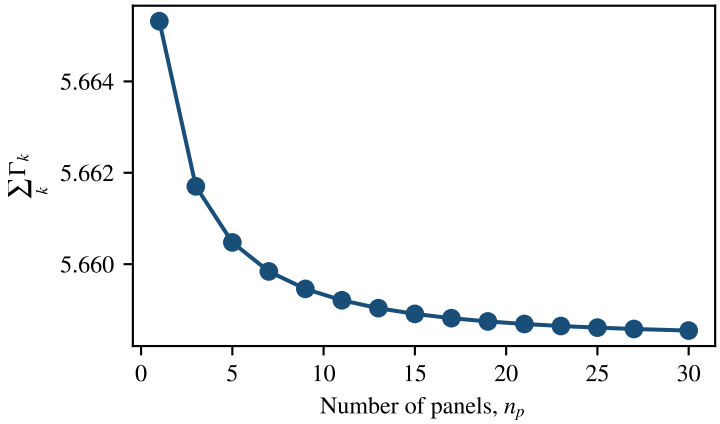


Figure 3.2: Convergnce of total airfoil circulation for increasing number of blade panels representing the flate-plate airfoil, n_p.

Δt. Namely, Δt can be normalized by the frequency of the periodic kinematic velocity, $\vec{U} = \vec{U}(t) \mid \vec{U} = \vec{U} \cdot \vec{U} \sin(\omega t)$, resulting in $\Delta t\, \omega / 2\pi$; or by the normalized *wake vortex spacing*, resulting in $\Delta t\, U_\infty / c$. We concluded from a preliminary analysis that the wake vortex spacing, defined here as:

$$\text{Wake vortex spacing} \equiv \frac{\Delta t\, U_\infty}{c}, \tag{3.3}$$

is a much more conservative definition for the normalized time-step sizing as $U_\infty / c > \omega / 2\pi$ in the assessed benchmark cases; thus, we selected it for this study.

We show in Fig. 3.3 the convergence of the total airfoil circulation for decreasing values of wake vortex spacing. Notice that the convergence is slower than logarithmic convergence for decreasing time-step sizings. That is because the unsteady solver has to perform at least $(t_f / \Delta t \times n_p)^2$ operations at each time step, thus making the computations of fine time-discretizations demanding. Thus, for this study, $\Delta t U_\infty / c = 0.2$ is deemed sufficiently accurate in front of the high computational times necessary for the time-stepping independent results.

Having established the sizing of the normalized input parameters, we validate our model against an experimental observation of a flow past a plunging flat plate at a high reduced frequency k. Here, we define k as:

$$k \equiv \frac{\omega c}{2 U_\infty}, \tag{3.4}$$

where ω is the frequency, in radians per second, of the flat-plate airfoil motion, notice that in this study,

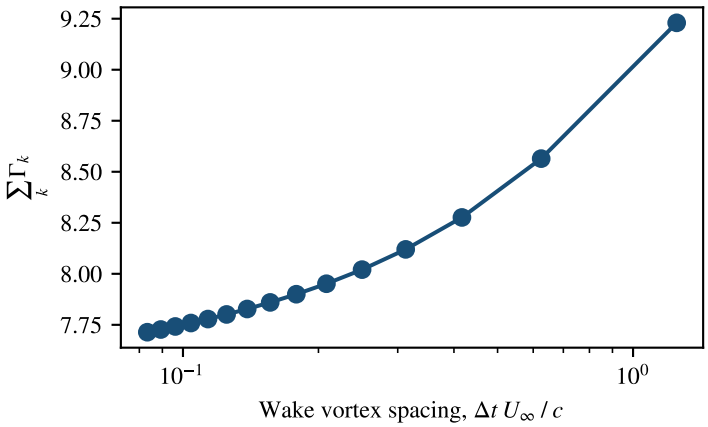


Figure 3.3: Convergnce of total airfoil circulation for increasing decreasing wake vortex spacing, defined here as $\Delta t\, U_\infty / c$.

both the assessed plunging and pitching motion profiles for the airfoil are sinusoidal functions of time; the above definition of k can always be used without loss to generality.

As discussed by Katz (2001) [2], we assume that the solution for lifting flows is unique and equal to that where the flow leaves smoothly at the airfoil's trailing edge. This assumption helps us fix the amount of circulation required in the flow. However, there is no obvious way of determining whether the Kutta condition is the best candidate. For instance, significant oscillation frequency, large amplitudes, and angles of attack can cause trailing-edge separation and thus violate the Kutta condition. Therefore, comparing and validating our discretization-independent model against experimental observations of highly unsteady flows is interesting. Here, we define such flows as those where $k > 0.1$.

In Fig. 3.4, we compare the vortex shed pattern of our model by overlaying an experimental flow visualization by [3]. The unsteady flow case shown in the figure consists of a sinusoidal-plunging motion profile at a plunging reduced frequency of $k = 8.5$ at $\alpha = 5$ degrees and $h/c = 0.019$, discretized at a wake vortex spacing of $\Delta t\, U_\infty / c = 0.009$. Notice that despite the high value of reduced frequency for the oscillating plunging motion, there is a high correlation between the experimental and numerical flows past the flat plates. Thus, the lift loadings from our numerical solution are also expected to agree well with experiments.

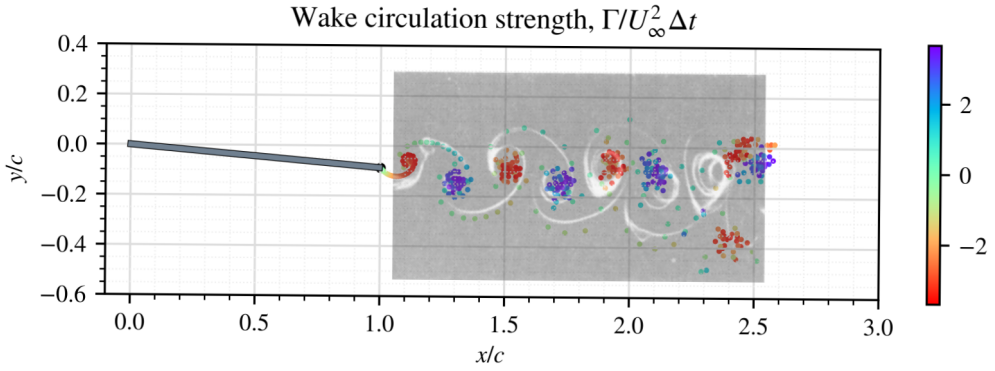


Figure 3.4: Flow past flat plate undergoing plunging motion at $k = 8.5$. Wake vortexes are colored with the normalized wake circulation strength. Experimental flow visualization from [3] are overlaid for reference.

We further extend our analysis to include vorticity, $\vec{\omega} = \omega_z \hat{k}$, and pressure coefficient,

$$c_p = 1 - \frac{\vec{u} \cdot \vec{u}}{U_\infty^2},$$

fields by superimposing the free-stream velocity field, U_∞, with the vortex-induced velocities obtained from computations of Eq. 2.2 for every lumped-vortex element in the flow. The vorticity and pressure-

coefficient fields for the current benchmark plunging flow are shown in Fig. 3.5. We also overlay the flow fields with the instantaneous streamlines.

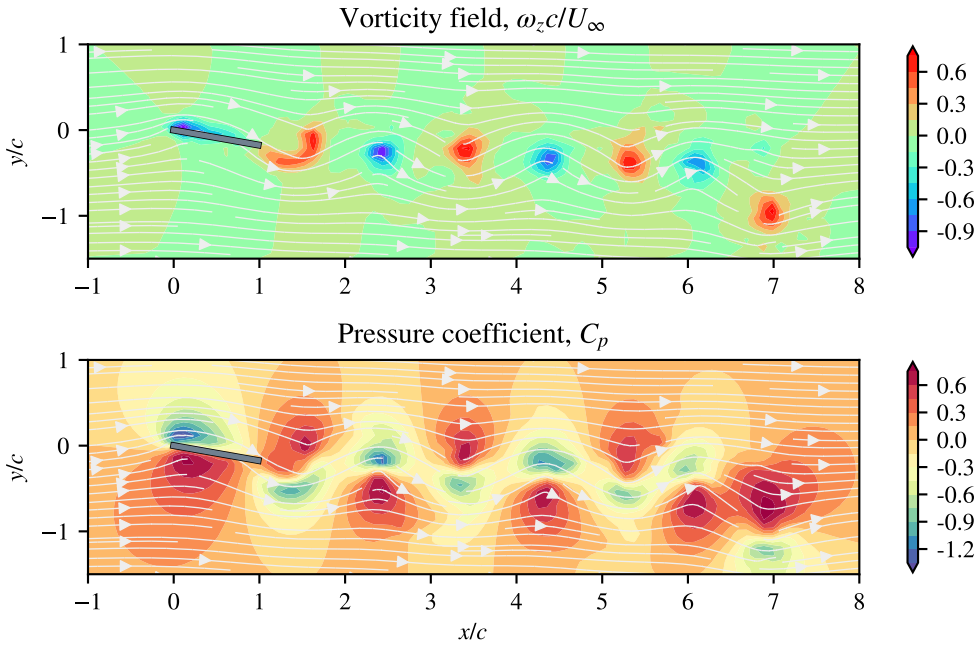


Figure 3.5: (Top) vorticity field and (bottom) pressure-coefficient field for the plunging flow at $k = 8.5$ case. The streamlines are overlaid in white for reference.

3.2. Steady flow cases

In this section, we test the discretization-independent model for steady flows at different angles of attack, α. The discretization is such that the flat-plate airfoil is modeled using $n_p = 10$ lifting plates. The wake length is of $U_\infty t_f/c = 25$, with a constant wake vortex spacing of $\Delta t\, U_\infty/c = 0.2$. The simulations in this subsection show that at $t = 0$, the airfoil is stationary relative to the inertial (fixed) reference frame. At $t > 0$, the airfoil is instantaneously accelerated to $\vec{U}$.

We show in Fig. 3.6 the lift per unit span,

$$L' = U_\infty \cos(\alpha) \sum_k \Gamma_k,$$

normalized by the free-stream dynamic pressure, $0.5U_\infty^2$. We also overlay the standard 2π-lifting curve from thin airfoil theory for reference. The forces are computed for eight different angles of attack ranging in $[0, 20]$ degrees. Notice also that, despite being referred to here as "steady results," the computational simulations carried out in this subsection still include vortex roll-up. However, since the airfoil does not accelerate for $t > 0$, and $t \gg 0$, the unsteady effects due to the shedding of the starting vortex are negligible, and the system can be considered steady-state.

Notice from Fig. 3.6 that the thin airfoil theory holds until $\alpha \approx 10$ degrees. For higher attitudes, the thin airfoil theory still predicts a linear relation between cross-flow forces and angle of attack. In contrast, the lifting panel method predicts an increase in lift production for increasing angles of attack. The main reason for such a difference is due to the small-angle-of-attack approximations of the thin airfoil model, where $\sin(\alpha) \sim \alpha$. The slight angle of attack approximations are no longer valid for larger angles of attack. Notice that for the lifting panel method, the lifting force increases with the cosine of the angle-of-attack, and thus both models diverge for $\alpha \gg 0$.

It is also essential to highlight that the implemented model can only predict fully-attached flows since the Kutta condition is reinforced by shedding vortex that leaves from the trailing edge. In realistic

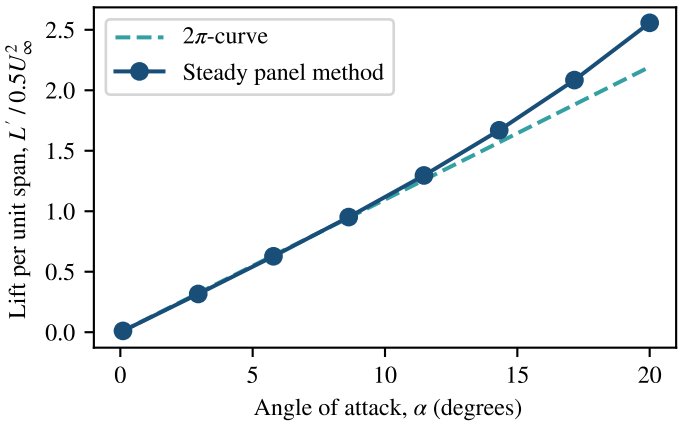


Figure 3.6: Lifting force (per unit span) normalized by free-stream dynamic pressure, $0.5U_\infty^2$, versus angle of attack. We overlay the 2π-cuve from the classical thin airfoil theory for reference.

flows, the flow over the flat plate would not have the energy to remain attached for high amplitude motions or angles of attack. Thus the fully-attached flow, and the Kutta condition itself, would not be respected.

3.3. Unsteady flow: oscillating plunging motion profile

After the steady flow results shown in the previous section, we discuss here the flows past plunging flat-plate airfoils at different reduced plunging frequencies, $k_{\dot{h}} = 0.02$, 0.05 and 0.1, such that:

$$k_{\dot{h}} \equiv \frac{\omega c}{2U_\infty} \mid \dot{h}(t,\omega) = 0.05 c\,\omega \sin(\omega t). \tag{3.5}$$

where $\dot{h} \equiv \partial h/\partial t$. Moreover, to avoid assessing cases for which the Kutta condition can not be guaranteed, we only consider plunging motion profiles with a small amplitude of $h/c = 0.05$. The airfoil is rotated at a fixed angle of $\alpha_0 = -5$ degrees relative to the x-direction in all assessed cases.

In Fig. 3.7, we show the lift coefficient for the first two oscillating motion periods, $t_f f_{\dot{h}} = 2$, for the three assessed reduced frequency values. We normalize the lift coefficient of the unsteady (plunging) motion cases by $2\pi\alpha$, corresponding to the lift force coefficient per unit span of a thin airfoil model in a steady flow. For the lowest tested reduced frequency motion, at $k_{\dot{h}} = 0.02$, the oscillating component of the lifting force is small in amplitude compared to the mean, "steady" value of $C_L/2\pi\alpha_0 = 1$. However, as we increase the reduced frequency of motion, the amplitude of the oscillating lift force component becomes more evident, revealing the unsteady nature of the plunging flows.

It is also interesting that higher motion frequencies yield larger phase gaps between the oscillating motion profile and lifting force. To understand this observation, we must first establish the sequence of events in the near-wake region during the oscillatory motion period. At $t = 0$, the airfoil and wake have circulation strengths of $\Gamma_{\text{airfoil}} = \Gamma_{\text{wake}} = 0$. At $t > 0$, the airfoil is instantaneously put into axial motion. At this moment, we impose that vortex-induced velocities must vanish the airfoil-normal velocity components, i.e., the circulation $\Gamma_{\text{airfoil}} \neq 0$. We also impose that the streamlines leave parallel to the airfoil's trailing edge (Kutta condition). From Kelvin's Theorem, such non-zero bound, airfoil circulation, Γ_{airfoil}, must necessarily vanish with the wake's circulation, Γ_{wake}, so that the total circulation in the flow is conserved. Equivalently, $\Gamma_{\text{airfoil}} = -\Gamma_{\text{wake}}$.

The combination of the non-entry boundary conditions with the Kutta condition is such that when an airfoil with positive α is put in axial motion, the flow over the suction (top) side accelerates relatively to the flow under the pressure (bottom) side, thus creating a net positive (clock-wise direction) circulation Γ_{airfoil}. Naturally, the starting vortex strength must be $-\Gamma_{\text{airfoil}}$, i.e., in the anti-clockwise direction. This anti-clockwise rotational motion in the first instants of the flow translates into a difference between

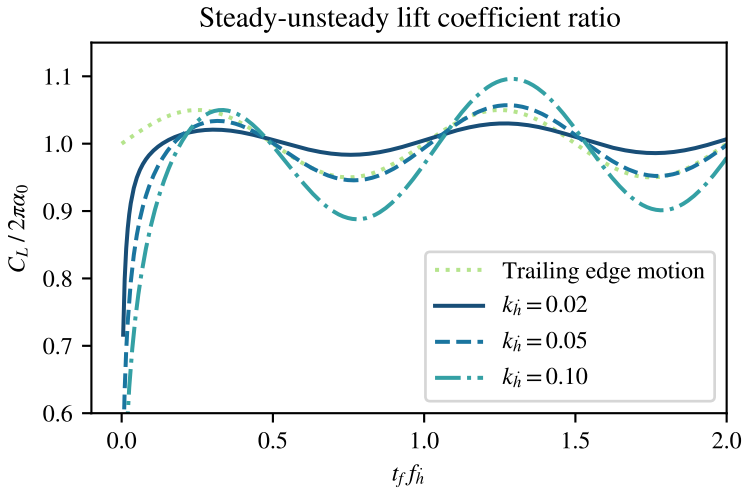


Figure 3.7: Lift force coefficient of plunging flows normalized by the lift force coefficient from thin airfoil theory, $2\pi\alpha$. Results shown for different reduced frequencies, k, as a function of the normalized simulation period, $t_f f_h = 2$.

$C_L(t = 0)/2\pi\alpha_0 = 0$ and $C_L(t \gg 0)/2\pi\alpha_0 = 1$ since the starting vortex is causing a downwash of the streamlines approaching the trailing edge, thus reducing the effective lifting force. However, as the starting vortex is advected downstream, C_L gradually increases to its steady-state value. This effect translates into the initial gradual increase observed ($t_f f_h < 0.25$) in C_L for all curves in Fig. 3.7.

In Fig. 3.8, we show the steady-state vorticity and pressure coefficient fields for the $k_h = 0.1$ case. Notice that, as expected, a suction (negative pressure coefficient) region is observed over the airfoil's upper surface. In contrast, net positive pressure coefficients are observed under the lower surface of the airfoil. From the Classical Mechanics point of view, this positive pressure gradient upwards is the cause of the positive lift force.

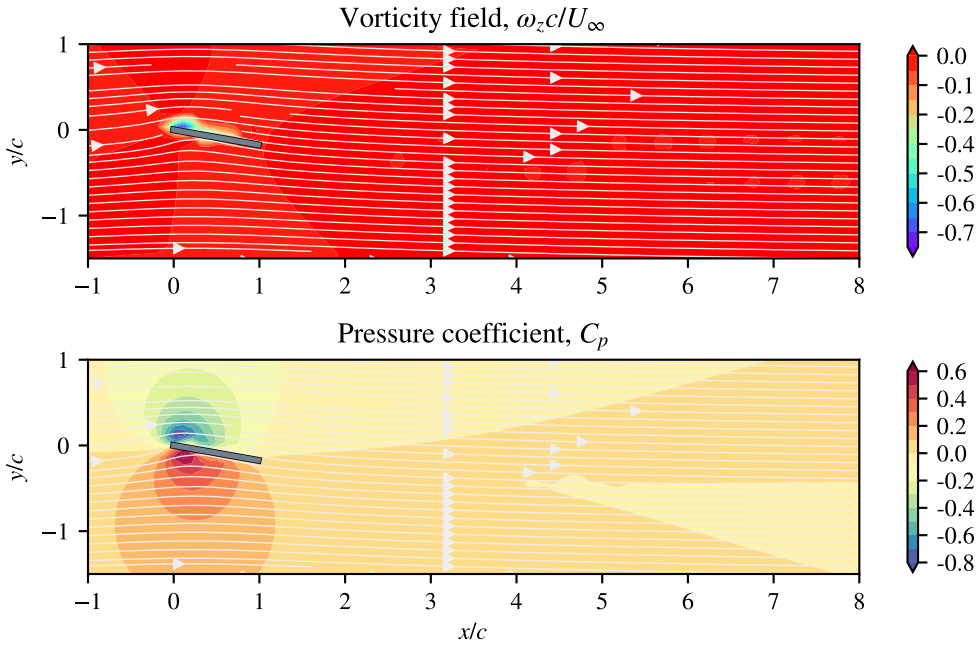


Figure 3.8: (Top) vorticity field and (bottom) pressure-coefficient field for the plunging flow at $k_h = 0.1$.

The vorticity field shown in Fig. 3.8 reveals that the steady-state flow is largely irrotational due to

the relatively low reduced frequency motion of the airfoil. However, there is negative, i.e., clock-wise-direction, vorticity over the airfoil, specially near the leading edge. These results are expected, since the circulation over the airfoil is positive, and larger in magnitude closer to the leading edge, where the downwash caused by the near-wake vortices is less pronounced.

3.4. Unsteady flow: oscillating pitching motion profiles

The vorticity field shown in Fig. 3.8 reveals that the steady-state flow is approximately irrotational due to the airfoil's relatively low reduced frequency motion. However, there is negative, i.e., clock-wise-direction, vorticity over the airfoil, especially near the leading edge. These results are expected since the circulation over the airfoil is positive and more significant in magnitude closer to the leading edge, where the downwash caused by the near-wake vortices is less pronounced.

$$\alpha(t, \omega) = \alpha_0 + \Delta\alpha \, \sin(\omega t),$$

where $\alpha_0 = 5$ degrees and $\Delta\alpha = 5$ degrees. The motion oscillating frequency, ω, is defined based on the reduced frequencies $k_{\dot{\alpha}}$, such that:

$$k_{\dot{\alpha}} \equiv \frac{\omega c}{2U_\infty} \mid \dot{\alpha}(t, \omega) = \Delta\alpha \, \omega \cos(\omega t). \tag{3.6}$$

In Fig. 3.9, we show lift coefficient results for oscillating pitching motion profiles at $k_{\dot{\alpha}} = 0.02$, 0.05 and 0.10. The lift force coefficients are normalized by $2\pi\alpha_0$, the lift force coefficient from the thin airfoil theory. From the results we show in the figure, it is evident that increasing values of reduced frequency yield in larger phase lag between the airfoil's motion and the lift force. Moreover, the maximum lift force is also reduced for increasing values of reduced frequency.

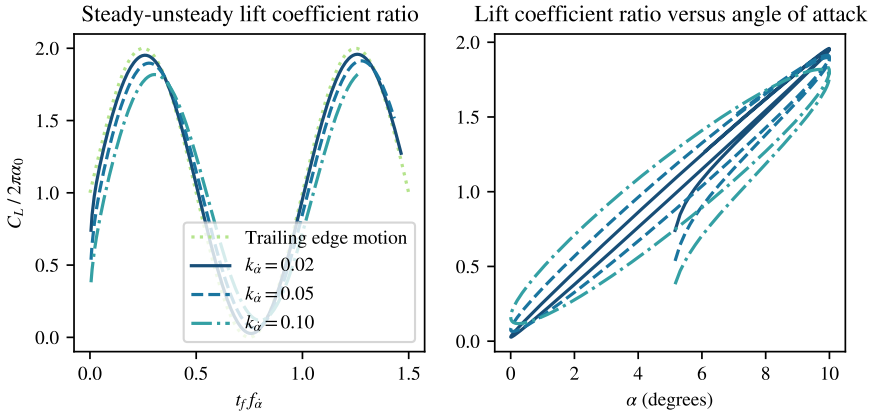


Figure 3.9: (Left) lift force coefficient of pitching flows normalized by the lift force coefficient from thin airfoil theory, $2\pi\alpha_0$. Results are shown for different reduced frequencies, k, as a function of the normalized simulation period, $t_f f_{\dot{h}} = 1.5$. (Right) lift coefficient versus angle of attack revealing hysteresis in the lift forces.

The phase lag between the airfoil's motion and lifting force comes from the fact that the wake vortexes are advected with the mean flow, i.e., with U_∞, whereas the airfoil motion oscillates with ω. For high values of $k_{\dot{\alpha}}$, the difference between the characteristics periods of airfoil motion and wake-vortex advection increases. For the continuous change of the angle of attack in a limited time, the change of the flow field lags behind the change of the angle of attack. For instance, when the angle of attack increases, the flow field at the larger angle of attack may still stay in the state of the flow field at the lower angle of attack, and thus the airfoil at the higher angle of attack reflects the aerodynamic state at the former state.

It is also interesting to notice that the above phase lag translates into a hysteresis loop, as seen in the right-hand side panel of Fig. 3.9. As can be seen, for the exact angle of attack, the airfoil produces higher lift during the upstroke than the downstroke motion. The change of lift force coefficient with

the angle of attack presents an anti-clockwise direction loop. When the airfoil performs the downstroke motion, the phase lag is such that the vortex wake is still building up a counterclockwise rotation motion in the flow, thus reducing the perceived angle of attack. During the downstroke motion, the opposite effect takes place. Furthermore, it can be observed that the hysteresis loops are broadened for more significant reduced frequency,

In Fig. 3.10, we show the vorticity and pressure coefficient fields of the $k_{\dot{\alpha}} = 0.1$ case. Notice since $k_{\dot{\alpha}}$ is considerably lower than 1, the vorticity of the flow is not evident in the wake, and thus, the steady-state wake is largely irrotational, besides the starting vortex (which is present in the limits of the plots).

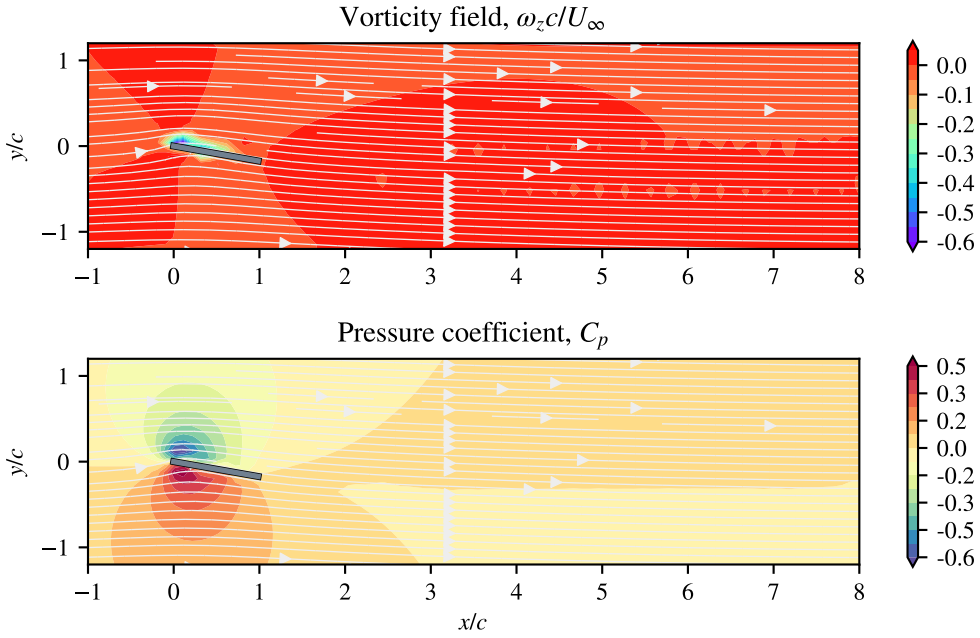


Figure 3.10: (Top) vorticity field and (bottom) pressure-coefficient field for the pitching flow at $k_{\dot{\alpha}} = 0.1$.

It is also interesting to compare the induction due to *shed vorticity* and *trailing vorticity*. From Kelvin's theorem, the circulation around a closed curve moving with the fluid remains constant. We also know from Helmholtz's Theorems that a vortex line cannot end in a fluid; it must extend to the boundaries of the fluid or must form a closed path. Let us now picture a three-dimensional, finite-span airfoil stationary to the potential flow at $t = 0$ and put on sudden forward motion at $t > 0$ and with a positive angle of attack α. If we reinforce the non-entry boundary condition and the Kutta condition, the strength of the vortex filament bound to the airfoil increases from zero to Γ_{airfoil}. From Kelvin's theorem, this increase on bound circulation must be compensated by shedding a vortex filament in the wake with strength $\Gamma_{\text{wake}} = -\Gamma_{\text{airfoil}}$. However, in a three-dimensional domain, Helmholtz's theorem requires that this (initial) shed vortex filament forms a closed loop with the bound vortex filament on the airfoil. Thus, two vortex filaments connecting the bound and initial shed vortex filament must form. We illustrate this scenario schematically in Fig. 3.11. These four connected vortex filaments form a *vortex ring*, with equal vortex strength Γ.

In the case of the airfoil put in a sudden forward motion described above, the bound vorticity is in the clockwise direction. Thus, the shed vorticity is in the anticlockwise direction. As this shed vorticity is advected downstream by the surrounding flow, both the bound and initial shed vortexes contribute to the vortex roll-up and the consequent downwash of the wake forming behind the airfoil, as illustrated in Fig. 3.11. In a three-dimensional flow, where the two trailing edges form between the bound and initial shed vortex, the trailing vortex filaments would be roughly parallel to the streamwise direction. Thus two extra components to downwash would form behind the blade, as illustrated in Fig. 3.12. Here, only the first (two-dimensional) effect due to the shed vortexes is modeled.

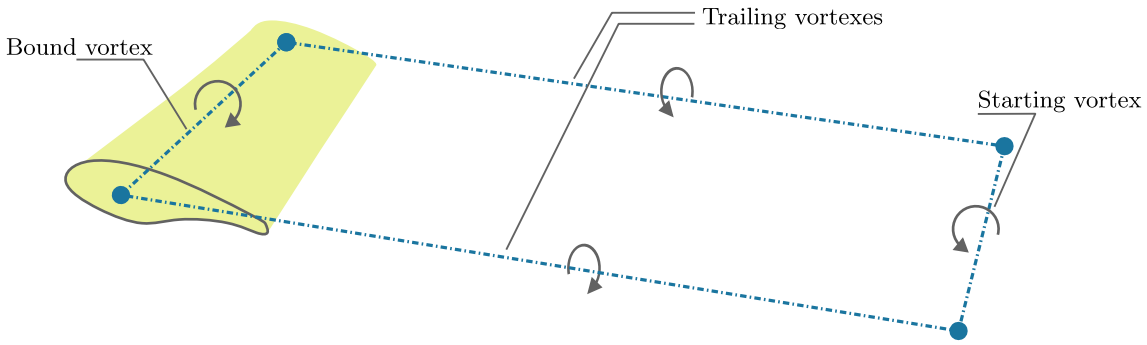


Figure 3.11: Schematic diagram illustrating bound, trailing and starting vortexes behind a three-dimensional, finite blade moving to the left-hand side of the page.

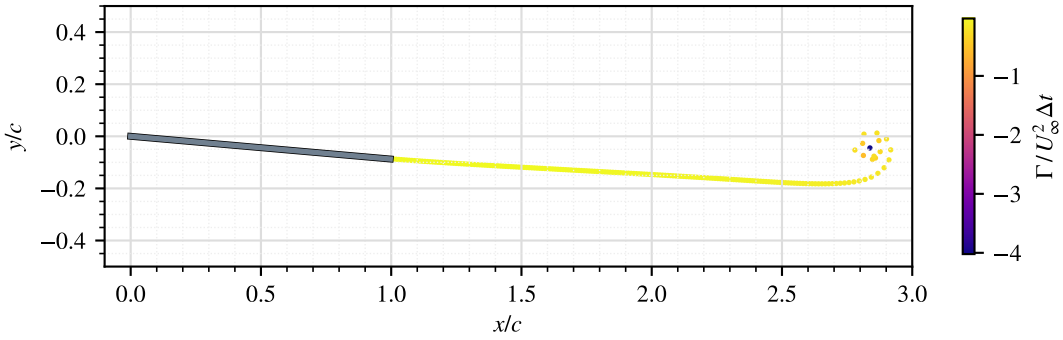


Figure 3.12: Shed wake vortex behind an airfoil moving at a constant speed of $\vec{U} = -U_x \hat{i}$. Notice the vortex roll-up around the starting (right-most vortex) wake vortex.

In this report, we have modeled the unsteady aerodynamics of an airfoil in a two-dimensional space. Thus, the bounded vortex filaments are simplified to one or more vortex points (i.e., we lump the singularity elements into points). In contrast, the trailing vortexes can not be defined as the flow is two-dimensional. Therefore, the extra induction and downwash caused by blade-tip vortices can not be modeled. Nonetheless, the current unsteady solver accounts for both vortex shedding and wake roll-up effects, missing in the steady, three-dimensional solvers of the "Frozen-Wake" methodology. Notice also that in a frozen-wake methodology, the two shed vortex shed at two consecutive time steps will have the same vortex strength as the flow is ideally steady-state. This last property makes the vortex strengths of two consecutive vortex rings of the frozen-wake methodology vanish at shed vortex locations. Thus, in a frozen wake methodology, only the strengths of the trailing vorticities contribute to the net induction. In summary: in the two-dimensional vortex panel method, the induction comes from the shed vorticity, whereas in a three-dimensional frozen-wake methodology, the wake-induction is a by-product of the trailing edge vorticity.

Notice that a more realistic unsteady airfoil model must account for boundary layer separation. For large angles of attack or high cross-flow accelerations, the streamlines coming from the leading edge may not have sufficient energy to encompass the adverse pressure gradient over the pressure side of the airfoil and thus can not sustain an attached flow. Moreover, vorticity is also formed at the leading edge for high angles of attack due to the vertical velocity gradient from the separated flow. Since, in the current methodology, the boundary layer is modeled as fully attached, both flow separation and leading edge vortex formations can not be accounted for. Therefore, adding flow separation modeling capabilities is a fundamental next step to the model benchmarked in this report. Moreover, we adopted the Kutta condition as a boundary condition to solve the linear vortex shedding model in the potential flow extension. For highly unsteady flows, the Kutta condition does not necessarily hold (non-lifting flows), and an alternative methodology that would allow for selecting different locations for the stagnation point is necessary.

4
Conclusions

In this report, we discussed and implemented a lifting panel method for two-dimensional unsteady flows around flat-plate airfoils. The implemented code accounts for shed vorticity via a Kutta-condition-inspired boundary condition for vortex shedding. Kelvin's theorem for circulation conservation is applied at each time step, and the circulation strength of bound and wake vortexes are determined by reinforcing the Kutta and non-entry boundary conditions. In this study, we explored steady flows at different angles of attack; plunging and pitching flows at different reduced frequencies; and discussed the differences and limitations of the implemented technique compared to a steady-state, "frozen-wake" methodology. We observed for the pitching cases that a hysteresis loop takes place for the lifting force. The hysteresis is explained by the inertia in the unsteady flow, such that the airfoil faces the downwash from the previously-shed vortexes that have not been shed sufficiently far from the airfoil. This extra downwash caused by the shed vorticity causes an increase in a lift during the pitching-down motion and a decrease in a lift during the pitching-up motion, thus forming a counterclockwise direction hysteresis loop. The effect is more significant for higher reduced frequencies.

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